ABSTRACT

SupplyGuard

An AI-Powered Shipment Delay Prediction and Risk Analysis System

Shipment delays in modern supply chains increase operational costs, reduce customer satisfaction, and disrupt logistics planning. This project proposes SupplyGuard, a Machine Learning-based web application designed to predict shipment delay risk before delivery, enabling logistics managers to identify high-risk shipments, make proactive decisions, and improve supply chain efficiency. The proposed system uses the DataCo Smart Supply Chain Dataset, which contains over 180,000 historical shipment records.

The dataset is preprocessed through data cleaning, feature selection, and transformation to identify relevant information for shipment delay prediction. Random Forest and Support Vector Machine (SVM) algorithms are employed to classify shipments according to their risk of late delivery, with the most suitable model integrated into the web application. The application is developed using Python Flask, SQLite, and HTML5, CSS3. It provides a web-based interface for shipment risk prediction, risk analysis, data visualization.

Dataset: https://data.mendeley.com/datasets/8gx2fvg2k6/5

Technology Stack :

Frontend: HTML5, CSS3, JavaScript, Bootstrap

Backend: Python, Flask

ML: Random Forest, Support Vector Machine (SVM)

Tools: Visual Studio Code, Jupyter Notebook, Git, GitHub

References:

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[2] N. Rezki and M. Mansouri, "Integrating Machine Learning and Deep Learning for Enhanced Supplier Risk Prediction," Acta Logistica, vol. 11, no. 3, pp. 387–396, 2024.
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